Lecture 1: Introduction

Statistical Foundations of Machine Learning

Feb 4, 2026

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Course logistics

- *Instructors*: Nikhil Karamchandani (nikhilk@ee.iitb.ac.in) and D. Manjunath (dmanju@ee.iitb.ac.in)

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  1. Homeworks: 15

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  1. Homeworks: 15
  2. Midsem: 35

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  2. Midsem: 35
  3. Endsems: 50

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Data Science: Probability and Statistics

- **Intersection:** Data Science lies at the meeting point of Statistics, Probability, and Computation.

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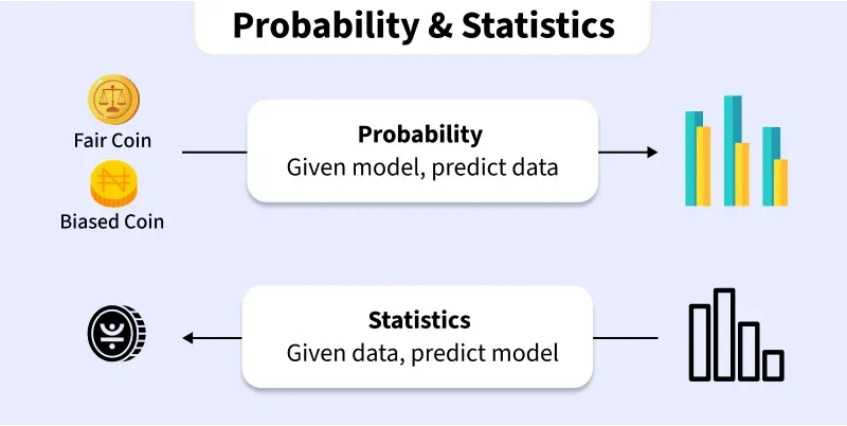

Illustration adapted from GeeksforGeeks:

https://www.geeksforgeeks.org/maths/probability-and-statistics/

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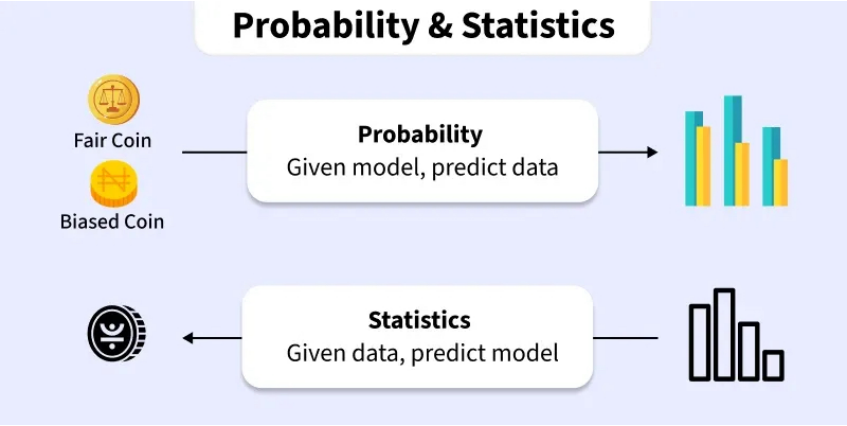
- **Statistics (Backward engine):**

$$P(\text{Model} \mid \text{Data})$$

Inferring truth from past observations.



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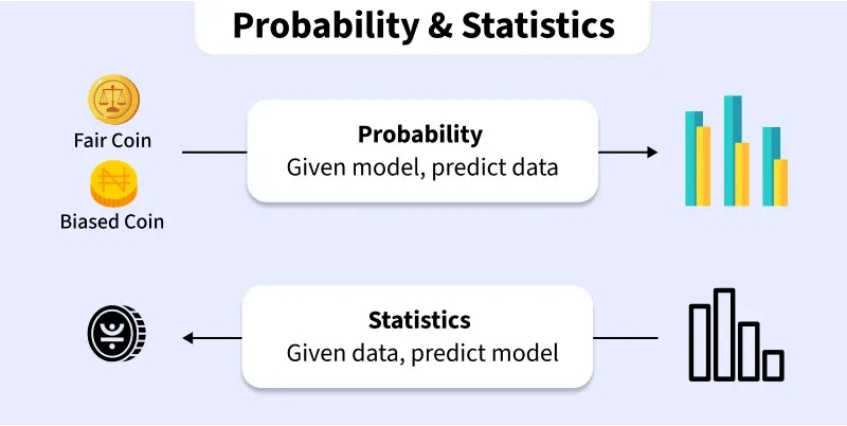
- **Probability (Forward engine):**

$$P(\text{Data} \mid \text{Model})$$

Predicting future outcomes from assumptions.



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Case 1: The "New-to-Credit" (NTC) Problem

Problem: A fintech app needs to set Rs 10,000 daily credit limit for a user with no credit history.

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Data: 10 days of UPI transaction logs to estimate daily liquidity.

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Day	1	2	3	4	5	6	7	8	9	10
Spend	400	350	500	420	**8,500**	380	410	440	390	460

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Observation: Most days are stable; Day 5 is a massive outlier.

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Descriptive Stats: Mean vs. Median

- **Arithmetic Mean ($\bar{x}$):**

$$\bar{x} = \frac{\sum x_i}{10} = \frac{12250}{10} = \mathbf{1225}$$

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Descriptive Stats: Mean vs. Median

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- **Median ($\tilde{x}$):** Sorted middle value.

$$\text{Sorted: } [350, 380, 390, 400, \mathbf{410}, \mathbf{420}, 440, 460, 500, 8500]$$

$$\tilde{x} = \frac{410 + 420}{2} = \mathbf{415}$$

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- **Insight:** The Mean is inflated by the outlier. The Median better represents the "true signal" of daily spend.

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Sensitivity of the Mean

Question: If 10% of data is "noise," how much does the mean lie?

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Question: If 10% of data is "noise," how much does the mean lie?
Model: 90% chance of Rs 500; 10% chance of Rs 50,000.

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$$E[X] = (0.90 \times 500) + (0.10 \times 50,000)$$

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$$E[X] = 450 + 5,000 = \mathbf{5450}$$

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Verdict: The Mean is physically pulled by the magnitude of rare events. It fails to represent the majority of days.

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Robustness of the Median

Logic: To pull the Median to 50,000, we need ≥ 6 outliers in 10 days.

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$$P(X \geq 6) = \sum_{k=6}^{10} \binom{10}{k} (0.1)^k (0.9)^{10-k}$$

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Calculation:

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Verdict: Median follows frequency, not magnitude. Better choice as the behavioral baseline for this user.

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Modeling Future Risk: Parameter Estimation

- **Next Step:** Having fixed a robust baseline (Median $= 415$), we now model the risk of extreme spend days.

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- Using our empirical Median value ($\tilde{x} = 415$):

$$\lambda = \frac{0.693}{415} \approx \mathbf{0.00167}$$

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Probability of "Stress"

Test: Approve loan if there is a $< 5\%$ chance of a spend spike $>$ Rs 2000.

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$$P(X > 2000) = \int_{2000}^{\infty} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_{2000}^{\infty}$$

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$$P(X > 2000) = e^{-0.00167 \times 2000} = e^{-3.34} \approx \mathbf{0.035}$$

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Decision: Risk (3.5%) is below our 5% threshold. **Loan Approved.**

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Case 2: A/B Testing

- **Problem:** Marketing wants to increase conversion rate ($\hat{p}$).

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- Group A (G_A): 300 clicks $\implies \hat{p}_A = \mathbf{0.30}$
- Group B (G_B): 340 clicks $\implies \hat{p}_B = \mathbf{0.34}$
- **Question:** Are the observed differences in $\hat{p}_A$ and $\hat{p}_B$ real, or due to random sampling noise?

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Probabilistic Model and CLT

- For each user i, define

$$X_i = \begin{cases} 1 & \text{if user clicks} \\ 0 & \text{otherwise} \end{cases} \quad \Rightarrow \quad X_i \sim \text{Bernoulli}(p)$$

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$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

- By the Central Limit Theorem (for large n):

$$\hat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$

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From Gaussian Fluctuations to Confidence

- From the previous slide:

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$$\hat{p} \approx \mathcal{N}\left(p,\ \frac{p(1-p)}{n}\right)$$

- Standardize the deviation:

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \ \approx\ \mathcal{N}(0,1)$$

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- Question: how large can random fluctuations be?
- For a standard normal variable:

$$\Pr(|Z| \leq 1.96) \approx 0.95$$

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- Interpretation:

$$\hat{p} \text{ typically lies within } \pm 1.96\sqrt{\frac{p(1-p)}{n}} \text{ of } p$$

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Calculating Reliability (Confidence Intervals)

From CLT: $\hat{p} \approx \mathcal{N}\left(p,\ \frac{p(1-p)}{n}\right)$, so randomness in clicks translates into uncertainty in $\hat{p}$.

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Standard Error (SE):

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$$SE_A = \sqrt{\frac{0.30 \cdot 0.70}{1000}} \approx \mathbf{0.0145}, \quad SE_B = \sqrt{\frac{0.34 \cdot 0.66}{1000}} \approx \mathbf{0.0150}$$

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$$SE_A = \sqrt{\frac{0.30 \cdot 0.70}{1000}} \approx \mathbf{0.0145}, \quad SE_B = \sqrt{\frac{0.34 \cdot 0.66}{1000}} \approx \mathbf{0.0150}$$

95% Confidence Intervals: Using $Z \approx 2$, we take $\hat{p} \pm 2SE$.

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Calculating Reliability (Confidence Intervals)

From CLT: $\hat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$, so randomness in clicks translates into uncertainty in $\hat{p}$.

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$$CI_A = 0.30 \pm 0.029 = [\mathbf{27.1}\%, \mathbf{32.9}\%]$$

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Conflict: Intervals overlap. The observed lift may be due to randomness.

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Closing the Loop: The Power of n

Key insight from CLT: $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$. Increasing n reduces randomness in $\hat{p}$.

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Closing the Loop: The Power of n

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Scenario: Increase sample size to **5,000** users per group.

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Scenario: Increase sample size to **5,000** users per group.

$$SE_B = \sqrt{\frac{0.34 \cdot 0.66}{5000}} \approx \mathbf{0.0067} \;\Rightarrow\; CI_B = [\mathbf{32.7\%}, \mathbf{35.3\%}]$$

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$$SE_A = \sqrt{\frac{0.30 \cdot 0.70}{5000}} \approx \mathbf{0.0065} \;\Rightarrow\; CI_A = [\mathbf{28.7}\%, \mathbf{31.3}\%]$$

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Decision: No overlap remains. The observed 4% lift is unlikely to be noise. **Deploy Gold.**

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Case 3: Finding Leading Indicators

Business Question: Which user behavior is more informative about future spend?

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Case 3: Finding Leading Indicators

Business Question: Which user behavior is more informative about future spend?

Candidates:

- In-App Session Time (X)
- Screen Brightness (Z)

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$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

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Goal: Distinguish meaningful signal from random noise.

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Comparative Correlation Table

Step 1: Center each variable by subtracting its mean.

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Comparative Correlation Table

Step 1: Center each variable by subtracting its mean.

Step 2: Multiply centered values to measure co-movement.

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User	X	Y	Z	$(x - \bar{x})$	$(y - \bar{y})$	Prod(XY)	$(z - \bar{z})$	Prod(ZY)
A	15	600	50	-11	-550	6,050	0	0
B	21	800	85	-5	-350	1,750	35	-12,250
C	25	1100	10	-1	-50	50	-40	2,000
D	31	1250	45	5	100	500	-5	-500
E	38	2000	60	12	850	10,200	10	8,500
Sum						**18,550**		**-2,250**

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E	38	2000	60	12	850	10,200	10	8,500
Sum						**18,550**		**-2,250**

Next: Normalize these sums to obtain correlation coefficients.

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Interpreting Signal vs. Noise

Correlation with Session Time (X):

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Interpreting Signal vs. Noise

Correlation with Session Time (X):

$$r_{XY} = \frac{18,550}{\sqrt{300 \times 1,125,000}} \approx \mathbf{0.71}$$

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Interpreting Signal vs. Noise

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$$r_{XY} = \frac{18,550}{\sqrt{300 \times 1,125,000}} \approx \mathbf{0.71}$$

Interpretation: Strong positive association — higher engagement aligns with higher spend.

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$$r_{ZY} = \frac{-2,250}{\sqrt{3250 \times 1,125,000}} \approx -\mathbf{0.03}$$

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Interpretation: No meaningful relationship — consistent with noise.

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Interpretation: No meaningful relationship — consistent with noise.

Conclusion: Session Time is a useful leading indicator; Screen Brightness is not.

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Conclusion: Statistics and Probability in Data Science

Core Theme: Data science is about making decisions under uncertainty.

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Conclusion: Statistics and Probability in Data Science

Core Theme: Data science is about making decisions under uncertainty.

Statistics helps us:

- Summarize noisy observations
- Identify robust patterns in data
- Separate signal from randomness

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Big Picture: Statistical thinking converts data into informed, defensible decisions.

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Big Picture: Statistical thinking converts data into informed, defensible decisions.

We will study these notions formally in this course!!

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Tentative lecture plan - first half

Lec1: Introduction
Lec 2: Probability axioms, conditional probability and independence,
Lec 3: Counting and sampling
Lec 4: Random Variables, CDF, Joint CDF, Independence
Lec 5: Discrete RVs (PMF), Joint / Conditional PMF
Lec 6: Continuous RVs (PDF), Joint / Conditional PDF
Lec 7: Expectation
Lec 8: Variance, Covariance and Correlation
Lec 9: Moment generating functions
Lec 10: Tail Inequalities and applications
Lec 11: Law of Large Numbers and Central Limit Theorem
Lec 12: Conditional expectation and MMSE

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References

Books:

- **A first course in probability, S. Ross**
- Probability, Random Variables and Stochastic Processes, Papoulis and Pillai
- Probability with Engineering Applications, Bruce Hajek, available at https://courses.grainger.illinois.edu/ece313/fa2020/probabilityAug21.pdf
- Probability and Random Processes, Grimett and Stirzaker
- One Thousand Exercises in Probability by Grimmett and Stirzaker.

Online resources:

- https://ocw.mit.edu/resources/res-6-012-introduction-to-probability-spring-2018/index.htm
- https://nptel.ac.in/syllabus/108106083/ (Probability foundations, Krishna Jagannathan, IITM)

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